UMA KitchenLab

The Human Candle

A professional learning guide for gravimetric combustion, power, energy cost, and basal metabolism

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Updated from live candle mass-loss data

Core Thesis

Mass loss is energy. Energy is power when divided by time. Power is cost when multiplied by time and price. The same arithmetic that measures a candle also explains a resting human body.

Measured result: one tea light in the captured run released about 39 W. Two similar tea lights would release about 78 W - a strong classroom analog for the basal metabolic power of one resting adult.

Quantity	Measured or reference value	Meaning
Measured mass loss	0.695 g in 12.50 min	Direct scale observation
Measured burn rate	0.0556 g/min	Fuel consumption rate
Energy released	29.2 kJ	Wax mass x heat of combustion
Measured tea-light power	38.9 W	Energy divided by elapsed time
Two similar tea lights	77.8 W	Near one resting adult

1. Executive Summary

The original UMA KitchenLab gravimetric principle is correct: a candle can be treated as a small combustion engine whose fuel mass decreases in time. If the mass-loss rate is measured, the heat release rate follows directly from the heat of combustion of the wax.

The important correction is not conceptual; it is experimental. The 70 W example is a valid nominal candle calculation for a candle burning around 0.100 g/min. The user-captured tea-light run burned more slowly, around 0.0556 g/min, and therefore produced about 39 W.

This makes the lesson better, not weaker. A tea light is a smaller, deliberately slow-burning candle. Two of the measured tea lights produce about 78 W, which is squarely in the range of resting human basal metabolic power.

Corrected teaching statement

A nominal candle burning paraffin at 0.100 g/min releases about 70 W. In the captured tea-light experiment, the measured burn rate was 0.0556 g/min, giving about 39 W. Two similar tea lights give about 78 W, an excellent physical analog for one resting adult.

Claim	Status after math check	Best wording
70 W candle	Valid as nominal case	A candle at 0.100 g/min is about 70 W.
Measured tea light	Lower because actual burn rate was lower	The captured tea light was about 39 W.
One candle equals one human	Ballpark for a larger/nominal candle	One nominal candle or two measured tea lights are close to one resting adult.
Two measured tea lights	Strongly supported	Two measured tea lights are about 78 W.

2. Learning Objectives

- Use a scale as a scientific instrument for combustion, not just weighing.

- Translate mass loss into energy using heat of combustion.

- Translate energy over time into power in watts.

- Compare candle power to household appliances, energy cost, and human metabolism.

- Use one bench experiment to connect MKS units, chemistry, thermodynamics, and measured-self physiology.

3. The Unified Idea: The Scale Measures the Fire

A burning candle appears to be a flame, but the measurable root of the event is mass flow. Paraffin wax is transformed into gaseous products that leave the local candle system. The scale records the departure as a decreasing mass.

Once the mass flow is known, the energy flow is known to first order. The heat of combustion converts grams of wax to joules. Dividing by elapsed seconds converts joules to watts. That is the whole gravimetric ladder.

Power = (mass loss rate) x (heat of combustion)

P = (dm/dt) x Hc

Observable	Instrument	Mathematical step	Meaning
Mass m	HX711/load cell eScale	Read grams	How much wax is present
Mass change dm	Scale over time	m_start - m_end	Fuel consumed
Burn rate dm/dt	Scale + clock	grams per minute	Fuel flow
Energy E	Calculation	dm x Hc	Chemical energy released
Power P	Calculation	E / time	Watts of heat release
Metabolic analogy	Comparison	W candle vs W body	Candle as visible resting metabolism

4. Central Visual Evidence: Bench and Browser

The photograph and dashboard are central to the proposition because they show the full chain from physical combustion to interpreted digital measurement. The flame is not isolated from the instrument; it is part of a complete evidence path.

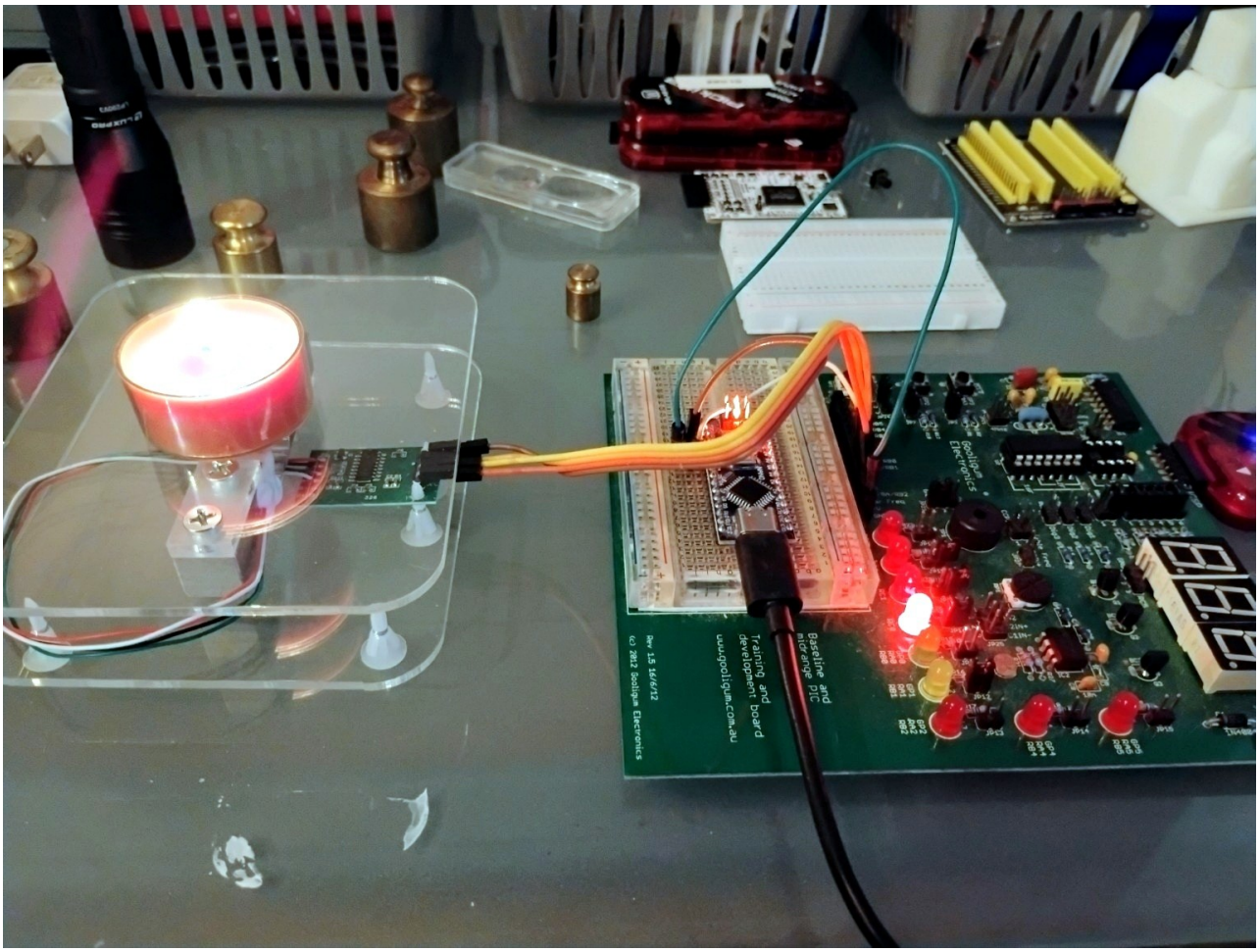


Figure 1. Physical UMA KitchenLab gravimetric candle-power bench. The tea-light candle sits on the load-cell platform; the HX711 sensing stage, PIC16F18424 training board, and Arduino Nano serial bridge complete the measurement chain.

Why this photograph matters

It makes the thesis concrete: a visible flame, a weighing platform, a microcontroller chain, and a browser instrument are one system. Combustion becomes measurable data.

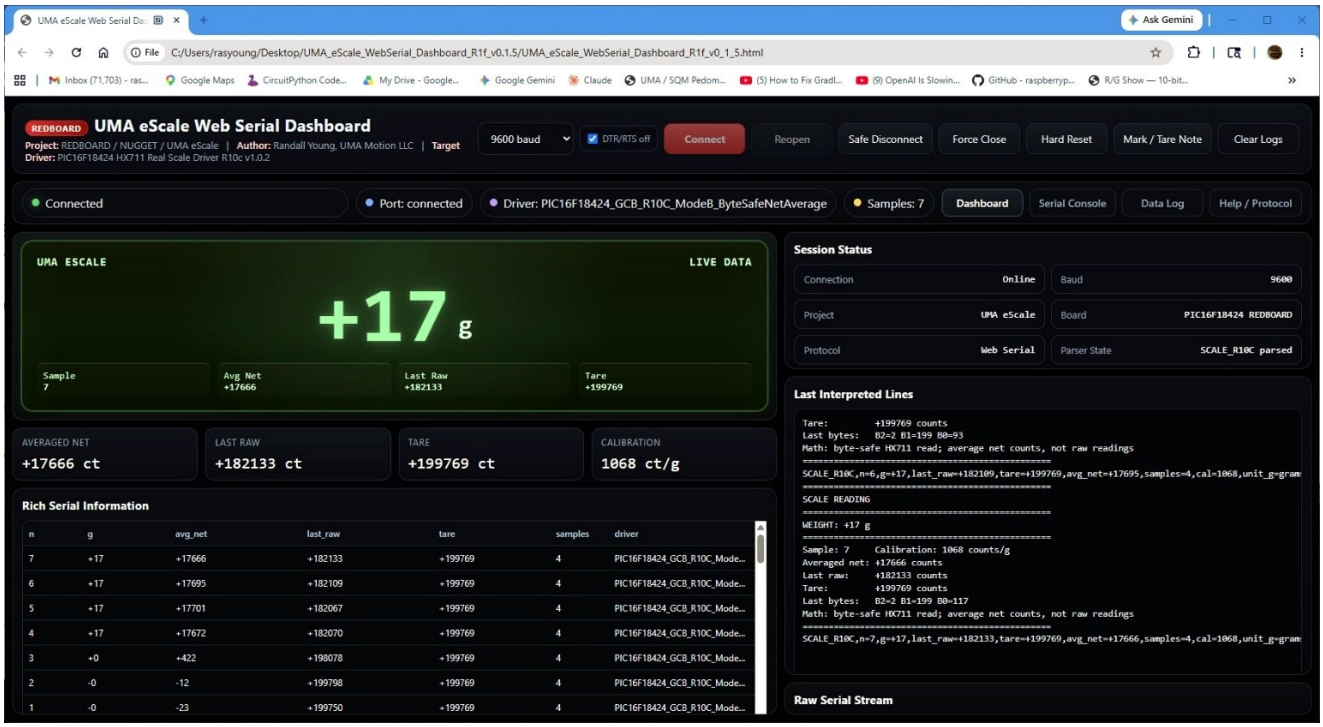


Figure 2. UMA eScale Web Serial Dashboard during live operation. The browser-based dashboard shows connection state, interpreted grams, raw counts, tare, calibration, and the serial evidence stream that supports the later energy and power calculation.

Together, these two images are the argument in visual form: the bench measures fuel mass, the dashboard records the measurement, and the learning guide converts that record into energy, power, cost, and metabolism.

5. Scientific Basis

5.1 Paraffin combustion

Paraffin candle wax is a mixture of straight-chain hydrocarbons. A representative formula such as C25H52 is good enough for classroom stoichiometry. During complete combustion, carbon and hydrogen combine with oxygen to form carbon dioxide and water vapor. Those products leave the candle region as gas, so the local mass on the scale declines.

C25H52 + 38 O2 -> 25 CO2 + 26 H2O(vapor)

For classroom power calculations, Hc = 42 kJ/g is a sound working value. A NIST-accessible candle-flame study reports literature and measured paraffin values in the same range, including net heat of combustion around 43 kJ/g and effective heat of combustion around 43.8 kJ/g. The 42 kJ/g value is therefore conservative and pedagogically clean.

5.2 Heat-release equation

The professional fire-science form of the calculation is the same as the KitchenLab form: heat release rate equals mass burning rate multiplied by heat of combustion, with combustion efficiency included when appropriate.

Qdot = efficiency x mdot x Hc

For a clean, small classroom candle calculation, using Hc directly gives the right order of magnitude. The important measured quantity is the mass burning rate.

6. User-Captured Candle Run

The usable burn segment was the long tare-stable segment of the captured candle data. The later reset/no-load dead-band segment is useful for instrument sanity checking, but not for the fuel-burn calculation.

Captured quantity	Value	How used
Start time	8:13:46 PM	Beginning of usable burn segment
End time	8:26:16 PM	End of usable burn segment
Elapsed time	12.50 min = 750 s	Power denominator
Start mass	14.634 g	Initial candle mass reading
End mass	13.939 g	Final candle mass reading
Mass lost	0.695 g	Fuel consumed
Burn rate	0.0556 g/min	Measured tea-light fuel flow

Note: This guide treats the endpoint calculation as the primary classroom value. A longer run with repeated samples would allow linear regression and uncertainty estimates.

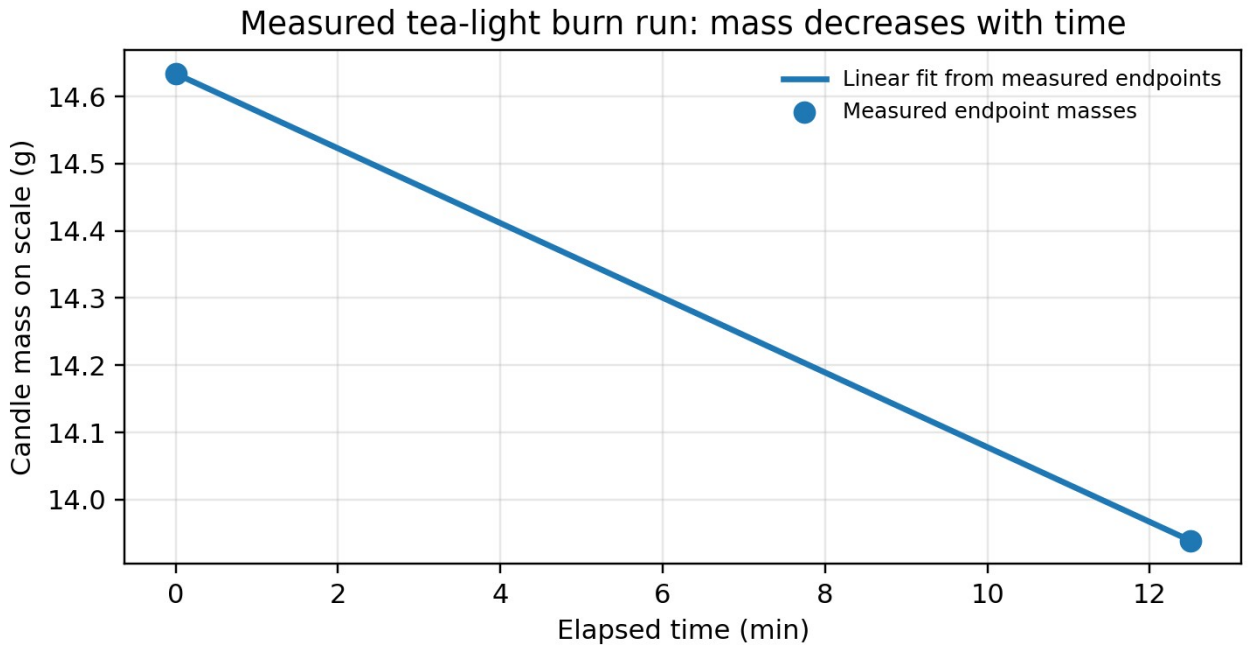


Figure 3. Endpoint-based mass-loss plot for the captured tea-light run. The line is a simple interpolation between the measured endpoints, not a replacement for a full time-series regression.

7. Calculation Walkthrough

$$\mathbf{dm = 14.634\ g - 13.939\ g = 0.695\ g}$$

$$\mathbf{dm/dt = 0.695\ g / 12.50\ min = 0.0556\ g/min}$$

$$\mathbf{E = 0.695\ g \times 42\ kJ/g = 29.2\ kJ}$$

$$\mathbf{P = 29.2\ kJ / 750\ s = 38.9\ W}$$

The measured candle did not fail the 70 W idea; it refined it. The specific tea light in the experiment burned at only about 56 percent of the nominal 0.100 g/min reference rate.

$$\mathbf{0.0556 / 0.100 = 0.56 = 56\%}$$

8. Why the Tea Light Was Lower Than 70 W

A tea light is a small, slow-burning candle. It has a small wick, a constrained melt pool, and a metal cup that helps regulate how much wax is melted and vaporized at one time. Since power is proportional to wax mass consumed per second, a lower burn rate directly gives a lower wattage.

The 70 W value should therefore be described as a nominal candle reference, not as the measured result for this particular tea-light run.

Case	Burn rate	Power using 42 kJ/g	Interpretation
Captured tea light	0.0556 g/min	38.9 W	Measured result
Two captured tea lights	0.1112 g/min	77.8 W	Resting-person analog
Nominal candle reference	0.1000 g/min	70.0 W	Valid general example

9. The Human Candle: Resting Metabolism Comparison

Human metabolism is also oxidation. Food molecules combine with oxygen; carbon dioxide and water leave the body; energy is released and used to maintain life. This does not make a human body a candle, but it makes the power comparison scientifically meaningful.

Reference	Calculation	Power
Average female BMR	1,410 kcal/day x 4184 J/kcal / 86400 s/day	68.3 W
Average male BMR	1,696 kcal/day x 4184 J/kcal / 86400 s/day	82.1 W
1 MET, 70 kg reference	70 kcal/hour x 4184 J/kcal / 3600 s/hour	81.4 W

Reference	Calculation	Power
Two measured tea lights	2 x 38.9 W	77.8 W

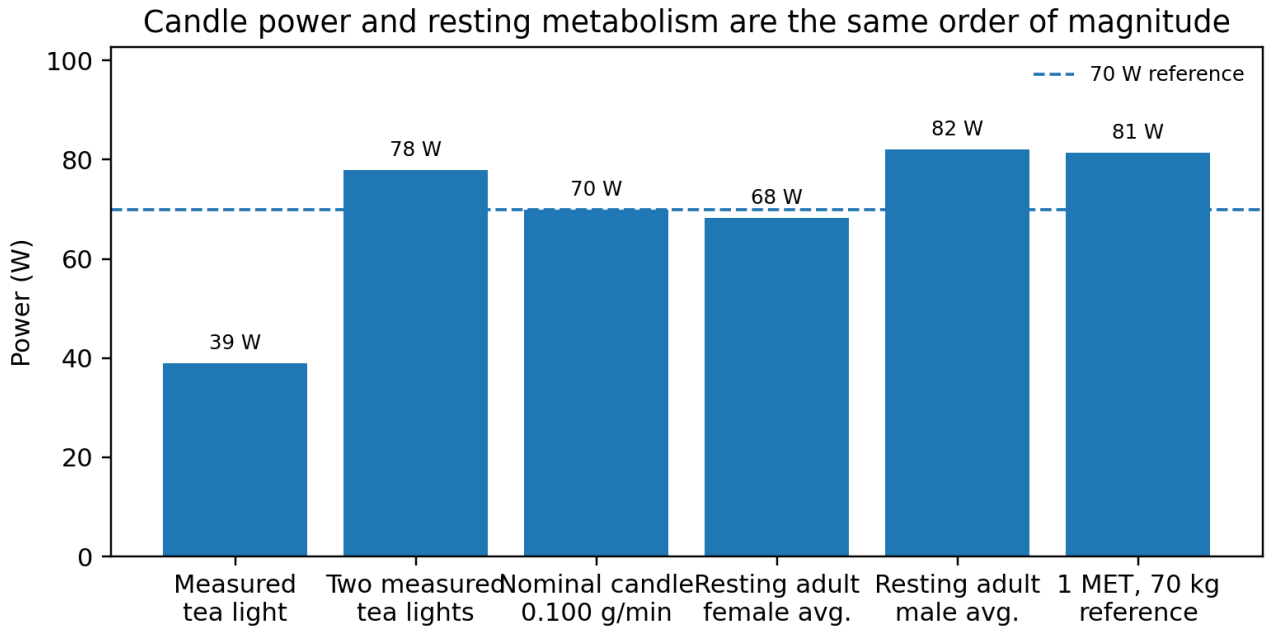


Figure 4. The two-tea-light result is not a forced analogy. It lands naturally in the same 70-90 W neighborhood as resting adult metabolism.

Best final classroom phrase

One measured tea light was about 39 W - roughly half of a resting adult. Two similar tea lights would be about 78 W, which is an excellent physical analog for the basal metabolic power of one resting adult.

10. Classroom Learning Sequence

Step	Student action	Core question	Mathematical idea
1	Place known mass on eScale	Can I trust the instrument?	Calibration and reference standards
2	Record candle mass before burn	What is the starting fuel mass?	Initial condition
3	Burn candle for timed interval	How does mass change with time?	Rate of change
4	Compute dm/dt	How fast is fuel leaving?	Slope
5	Multiply by heat of combustion	How much energy was released?	Unit conversion
6	Divide by elapsed time	What is the power?	Watts = joules/second
7	Compare with a person	How much power is a resting body?	Modeling by analogy
8	Compare with toaster/microwave/grid	What does energy cost?	Energy economics

11. Student Worksheet Core

1. Record start mass, end mass, and elapsed time.
2. Calculate mass lost: dm = m_start - m_end.
3. Calculate burn rate: dm/dt = mass lost / elapsed minutes.
4. Calculate chemical energy: E = dm x 42 kJ/g.
5. Calculate power: P = E / elapsed seconds.
6. Compare one candle, two candles, and a resting adult.
7. Write one sentence explaining why a smaller candle can have lower wattage.

12. Instrument Architecture and UMA Continuity

This experiment is not just a thermodynamics demonstration. It is a complete UMA signal chain: physical event -> sensor -> edge processor -> serial bridge -> browser dashboard -> interpreted model.

Stage	Hardware or role	Contribution
Sensor	Load cell + HX711	Converts force/mass into digital counts
Edge MCU	PIC16F18424 on Gooligum/Redboard-style board	Reads sensor and formats values
Bridge	Arduino Nano USB-Serial	Relays protocol to the computer
Transport	Chrome Web Serial	Browser-native acquisition path
Dashboard	UMA eScale LCD-style display	Shows grams and logs data
Model	Mass -> energy -> power	Turns observation into physical meaning

The same architecture can later support pop-can calorimetry, water-clock mass accumulation, battery-resistor heating, robot energy studies, and WalkWise/SQM+ score-oriented measurement. The common language is not the hardware; it is the chain from observable to model.

13. Extensions

Extension	Measurement	Teaching payoff
Pop-can calorimeter	Water temperature rise plus candle mass loss	Captured heat vs chemical energy; efficiency
Battery/resistor water bath	Voltage, current, time, water temperature	Electrical energy as controlled comparison
Water clock	Accumulated water mass vs time	Integration as accumulation
Flame flicker	Phototransistor frequency signal	Frequency counter and convection signature
Human power analogy	BMR/MET reference values	Measured World to Measured Self bridge

14. Safety, Limits, and Honest Claims

Open flame requires adult supervision, fire-safe surfaces, ventilation, and a method for extinguishing the candle. The scale and wiring should be protected from heat and wax.

The gravimetric calculation estimates chemical heat release. It does not say that all heat reaches a pot, can, or water sample. Calorimeter efficiency is a separate measurement.

The human comparison is a power-scale analogy. It is not a medical measurement and should not be presented as diagnostic. Human BMR varies with body size, composition, sex, age, temperature, illness, and other factors.

The captured tea-light result should be reported as about 39 W for one tea light and about 78 W for two similar tea lights. The 70 W candle remains a valid nominal candle reference.

Appendix A. Calculation Reference

Symbol	Meaning	Value used here
dm	Mass of wax consumed	0.695 g
dt	Elapsed time	12.50 min = 750 s
dm/dt	Mass burn rate	0.0556 g/min
Hc	Paraffin heat of combustion	42 kJ/g classroom value
E	Chemical energy released	29.2 kJ
P	Thermal power released	38.9 W

Appendix B. Recommended Replacement Text for the Original PDF

A candle burning paraffin at about 0.100 g/min releases roughly 70 W, which is in the same range as the basal metabolic power of a resting adult. In the captured UMA tea-light run, the measured burn rate was lower: 0.695 g over 12.50 min, or 0.0556 g/min. Using 42 kJ/g for paraffin, this gives 29.2 kJ over 750 s, or about 39 W. Therefore one measured tea light is roughly half of a resting adult, while two similar tea lights produce about 78 W - an excellent physical analog for one resting person.

Appendix C. References and Source Notes

- Young, R. S. UMA KitchenLab: Gravimetric Principles of Combustion & Energy Measurement. Draft PDF supplied by author, May 2026.

- Hamins, A. et al. Characterization of Candle Flames. NIST-accessible publication. Key values: paraffin net heat of combustion about 43.1-43.3 MJ/kg and effective heat of combustion about 43.8 MJ/kg; heat-release rate expressed as mass burning rate times heat of combustion with combustion efficiency.

- Cleveland Clinic. BMR (Basal Metabolic Rate): What It Is & How To Calculate It. Last updated Nov. 20, 2024. Cited values: average male BMR about 1,696 kcal/day; average female BMR about 1,410 kcal/day; BMR varies by body size, lean mass, sex, age, and other factors.

- 2018 Physical Activity Guidelines Advisory Committee Scientific Report, Appendix 1 Glossary. U.S. Department of Health and Human Services. MET definition: one MET is the rate of energy expenditure while sitting at rest, approximating oxygen uptake of 3.5 mL/kg/min for most people.

Appendix D. One-Page Summary for Students

The one-line lesson

If you can measure mass and time, you can measure energy and power.

Plain-English step	Formula	Captured value
How much wax disappeared?	dm = start - end	0.695 g

Plain-English step	Formula	Captured value
How fast did it disappear?	dm/dt	0.0556 g/min
How much energy was in that wax?	E = dm x 42 kJ/g	29.2 kJ
How powerful was the flame?	P = energy / seconds	38.9 W
What about two tea lights?	2P	77.8 W
Why compare with a person?	Resting adult: about 70-90 W	Same order of magnitude

Appendix E. Quick Teacher Script

1. The candle is not just glowing; it is consuming fuel at a measurable rate.

2. The scale sees the fuel leaving as gas. That is why the mass goes down.

3. A gram of paraffin contains about 42 kilojoules of chemical energy.

4. When we divide energy by seconds, we get watts. That makes a candle comparable to a toaster, a battery heater, or a human body at rest.

5. Our measured tea light was about 39 W. Two similar tea lights are about 78 W, which is very close to the power of a resting adult.

Final takeaway

The numbers are not magic. They come from mass, time, and one stored-energy constant. This is the KitchenLab version of seeing the invisible through measurement.